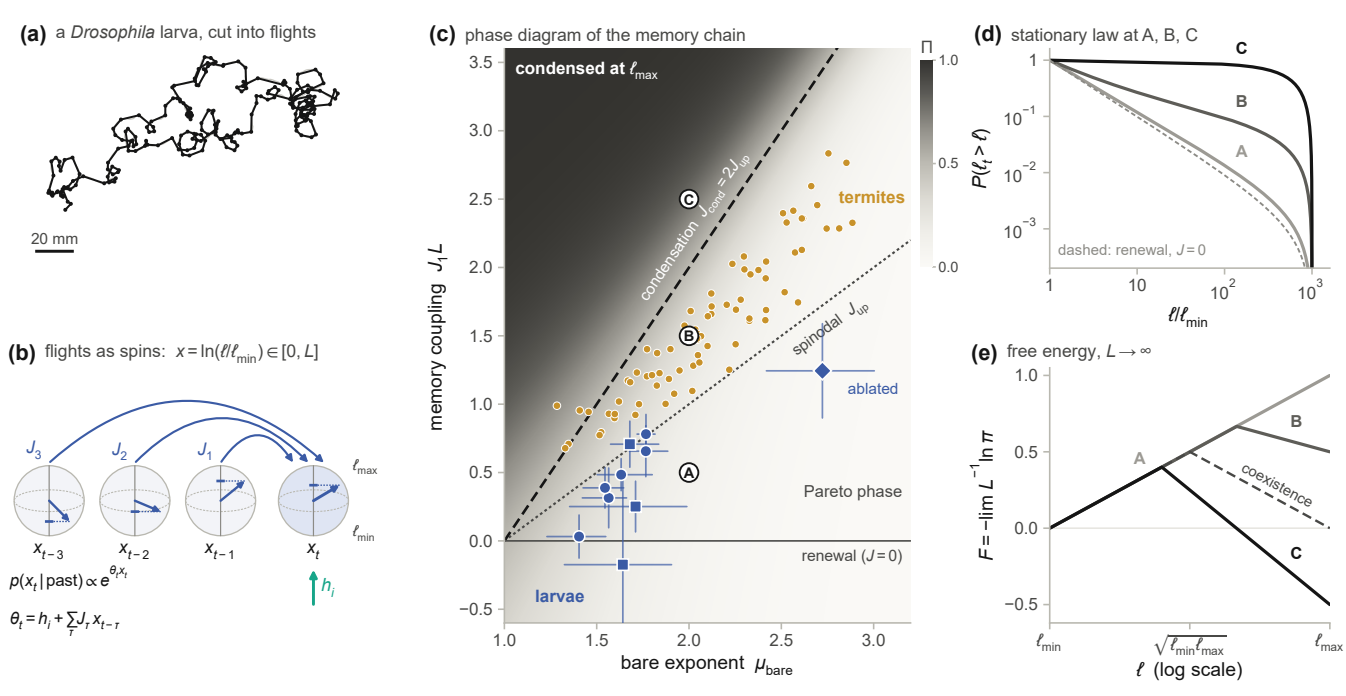

(a) a Drosophila larva, cut into flights
20 mm
(b) flights as spins: $x = \ln(\ell/\ell_{\min}) \in [0, L]$
J_3
J_2
J_1
$\ell_{\max}$
$\ell_{\min}$
x_{t-3}
x_{t-2}
x_{t-1}
x_t
$p(x_t \mid \text{past}) \propto e^{\theta_t x_t}$
h_i
$\theta_t = h_i + \sum_\tau J_\tau x_{t-\tau}$
(c) phase diagram of the memory chain
condensed at $\ell_{\max}$
condensation $J_{\text{cond}} = 2J_{\text{up}}$
termites
spinodal J_{up}
ablated
Pareto phase
renewal ($J = 0$)
larvae
memory coupling $J_1 L$
bare exponent μ_{bare}
Π
1.0
0.5
0.0
(d) stationary law at A, B, C
C
B
A
dashed: renewal, $J = 0$
$P(\ell_t > \ell)$
$\ell/\ell_{\min}$
(e) free energy, $L \to \infty$
A
B
C
coexistence
$F = -\lim L^{-1} \ln \pi$
$\ell_{\min}$
$\sqrt{\ell_{\min}\ell_{\max}}$
$\ell_{\max}$
ℓ (log scale)